

Anson Chen

206-532-8199 | ac283@cs.washington.edu | [linkedin.com/in/anson-c-chen](https://www.linkedin.com/in/anson-c-chen) | github.com/ansoncchen | US Citizen

EDUCATION

University of Washington

Seattle, WA

Bachelor of Science in Computer Science, GPA: 3.84

Expected June 2028

- **Relevant Coursework:** Data Structures/Parallelism, Deep Learning, C/Assembly, Linear Algebra

EXPERIENCE

Amazon Web Services

June 2026 – September 2026

Incoming SDE Intern, Compute Optimization Model Team

Seattle, WA

UbiComp Lab, UW Allen School

Seattle, WA

Undergraduate Research Assistant

April 2026 – Present

- Built automated video analysis pipeline using **Segment Anything Model 3** instance segmentation to detect, track, and quantify object interactions across 70+ hours of underwater footage for marine biology research.
- Developing stochastic simulation engine with biologically-informed random walks and **CMA-ES** optimization, evaluating physical design parameters across thousands of configurations to inform hardware prototyping.

ICTD Lab, UW Allen School

Seattle, WA

Undergraduate Research Assistant

February 2026 – Present

- Built multimodal ML pipeline aligning pretrained vision and geospatial foundation model embeddings over 100k+ camera trap images in **PyTorch**, achieving **0.849** cosine similarity via Ridge regression across three ecological benchmarks.
- Co-developing edge-deployable mobile vision classifier that fuses learned geographic embeddings with image features via multimodal conditioning, evaluated against vision-only baselines and targeting sub-50ms on-device inference.

Flow Z-Pinch Lab, University of Washington

Seattle, WA

Undergraduate Research Assistant

February 2026 – May 2026

- Rebuilt legacy PHP 4 fusion experiment logbook as a **FastAPI** REST API with JWT authentication and **Pydantic** validation, paired with a **SvelteKit** frontend, deployed behind **nginx**.

PROJECTS

DATect: Harmful Algal Bloom Forecasting | *Python, Sklearn, PyTorch, FastAPI, GCP*

2024 – 2026

- Created the first quantitative forecast for toxic algal events in the Pacific Northwest; **XGBoost + Random Forest** ensemble with per-site hyperparameter tuning across 10 sites achieves **$R^2=0.315$** on temporal holdout and **0.859** spike recall, a **434%** improvement over persistence baseline ([Github](#)).
- Designed and trained an unsupervised **3D convolutional autoencoder** over a 22-year, 4-channel satellite imagery cube (~4,700 composites) with masked-autoencoder pretraining; only method with positive R^2 at 7-day forecast lead across all Pacific Northwest regions.
- Engineered automated data pipeline using **Pandas** and **Polars** integrating 14 disparate sources including real-time NOAA satellite feeds; built full-stack deployment with **React** frontend and **FastAPI** backend containerized via **Docker** on **GCP**.

Codegraph: Codebase Knowledge Graph Platform | *Atlassian Forge, Gemini, React, Python*

Oct 2025

- Built a Forge-powered platform that parses GitHub repositories using **Gemini** agents to classify functions, detect cross-file dependencies, and render architectural flows as an interactive directed graph.
- Placed **Atlassian Title Track** at DubHacks, the largest collegiate hackathon in the Pacific Northwest.

AWARDS & HONORS

Grand Award, Regeneron ISEF

May 2024

Received 2nd Award in the largest ISEF category, selected from 100,000+ projects worldwide.

Regeneron Science Talent Search Scholar

Jan 2025

One of six Washington State scholars chosen nationally as a Top 300 Young Scientist.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL, Bash, HTML/CSS

Frameworks: React, Node.js, SvelteKit, FastAPI, Flask, PyTorch

Developer Tools: Git, Docker, Kubernetes, AWS, Google Cloud Platform, PostgreSQL, Redis, nginx, Linux, CI/CD

Libraries: Pandas, Polars, NumPy, Scikit-learn, XGBoost, Pydantic, Matplotlib